

Machine Learning A (2026–2027)

Home Assignment 3

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1 Optimal Classification (16 points)

1.1 Question 1

Let us map the original label space $Y \in \{0, 1\}$ to a new space $Y' \in \{-1, 1\}$ following the original probabilities table for X and Y .

We define our decision function based on the original probabilities:

$$\delta'(x) = P(Y' = 1 \mid X = x) - P(Y' = -1 \mid X = x)$$

The classification rule in the mapped space is:

$$h'(x) = \text{sgn}(\delta'(x))$$

Evaluating this for the given inputs:

$$\begin{cases} \delta'(0) = 0 - 1 = -1 \Rightarrow h'(0) = \text{sgn}(-1) = -1 & \text{if } x = 0 \\ \delta'(1) = 0.2 - 0.8 = -0.6 \Rightarrow h'(1) = \text{sgn}(-0.6) = -1 & \text{if } x = 1 \end{cases}$$
$$\Rightarrow h'(x) = -1$$

Mapping back to the original space Y , the optimal classification rule in the original space is $h(x) = 0$.

The expected risk (0-1 loss) is the probability of misclassification:

$$\begin{aligned} R &= P(h(X) \neq Y) \\ &= \sum_{x \in X} P(X = x) P(h(x) \neq Y \mid X = x) \\ &= \sum_{x \in X} P(X = x) P(Y = 0 \mid X = x) \\ &= 0.5 \cdot 0 + 0.5 \cdot 0.2 = 0.1 \end{aligned}$$

1.2 Question 2

In the case of $X = 0$, the probabilistic classifier still always predicts the correct label $Y = 0$, since $P(Y = 0 \mid X = 0) = 1$.

On the other hand, when considering $X = 1$ the expected risk is given by the probability of a mismatch between the classifier and the true label:

$$\begin{aligned} &P(h(x) \neq Y \mid X = 1) \\ &= P(h(x) = 0 \mid X = 1) P(Y = 1 \mid X = 1) + P(h(x) = 1 \mid X = 1) P(Y = 0 \mid X = 1) \\ &= 0.8 \cdot 0.2 + 0.2 \cdot 0.8 = 0.32 \end{aligned}$$

The total expected risk is therefore:

$$\begin{aligned} R &= P(h(X) \neq Y) \\ &= \sum_{x \in X} P(X = x) P(h(x) \neq Y \mid X = x) \\ &= 0.5 \cdot 0 + 0.5 \cdot 0.32 = 0.16 \end{aligned}$$

1.3 Question 3

No, it is not possible to find a better classifier than the best one previously found, that is the deterministic classifier defined in section 1.1. This is because by mapping the label space to $Y' = \{-1, 1\}$ the risk of this classifier corresponds to the Bayes risk, known to be the minimum over all possible measurable functions h .

$$h^{\text{Bayes}}(x) = \text{sgn}[\mathbb{E}_p(Y \mid X = x)]$$

2 Logistic Regression (50 points)

2.1 Cross-entropy error measure (16 points)

2.1.1 Rewriting negative logarithmic likelihood as cross-entropy

2.1.2 Cross-entropy and logistic regression

2.2 Logistic regression loss gradient (24 points)

2.2.1 Gradient for -1 and 1

2.2.2 Gradient for 0 and 1

2.2.3 Influence of misclassified examples

2.3 Log-odds (10 points)

3 Sleep Well (34 points)

3.1 Data understanding and preprocessing (4 points)

3.2 Classification (30 points)

3.2.1 Logistic regression

3.2.2 Random forest

3.2.3 Nearest neighbor

4 Overfitting (optional, 0 points)